

DATA PAPER

Multiyear genotype characterization of eastern spruce budworm outbreaking populations from Quebec and adjacent regions

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Abstract

Population outbreaks are characterized by irruptive changes in population density and connectivity resulting in rapid demographic and spatial expansion, often at the landscape scale. Outbreaks are common across multiple taxa, many of which inhabit northern ecosystems. Outbreaks of Lepidopteran defoliators in forest ecosystems are a particularly compelling example of this phenomenon, given the massive spatial scales over which these outbreaks can occur, their frequency, and socioeconomic impacts. The eastern spruce budworm (SBW) is a native outbreaking Lepidopteran defoliator of North American boreal forests. Cyclic outbreaks of the SBW influence ecosystem functioning and resilience, as well as forest productivity, timber supply, and other socioeconomic values related to management and mitigation. Despite these significant impacts, the ecological and biological drivers and outcomes of these outbreaks remain poorly understood. Here, we present an extensive genotypic dataset for 1998 geo-referenced SBW individuals collected between the years of 2012 and 2017, during the rising and peak phases of an outbreak that began approximately in 2006. Our sampling covers an unprecedented scope in the extent and number of individuals collected between 2012 and 2017 from Quebec, and in 2015 from New Brunswick (Canada) and from Maine (USA), from multiple SBW life stages, including early and late instar larvae (L2–L6), pupae, and adult moths. Genomic DNA extraction was followed by library preparation and high-throughput sequencing using Genotyping-by-Sequencing (GBS). Samples were genotyped for single nucleotide polymorphisms (SNPs) and aligned to the bw6 version of the SBW genome. This dataset represents one of the most extensive genotypic datasets to date for a boreal insect and is unique as it includes multiple years during a developing (ongoing, at time of sampling) outbreak. Sampling effort covered areas close to the epicenter of the outbreak (Quebec/Canada) and adjacent

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areas affected by the outbreak progress. This dataset also provides genome-wide characterization of SBW populations from Quebec, serving as a standard for the identification of future samples regarding their locality of origin, structure and connectivity. These data represent a valuable novel resource for further study of the spatial and temporal dynamics of SBW, and how spatial genetic diversity and gene flow are affected by population outbreaks. These data provide a temporal snapshot of SBW genetic diversity, which can serve as baseline for future studies regarding outbreaks, and the impact of human-induced environmental changes on complex population dynamics. This genotype dataset comprises a unique representation of genomic-level composition and variation observed in subsequent generations of an irruptive, cyclic outbreaking species and is of utmost importance for exploring and describing how accelerated demographic variation impacts the development of spatial genetic structure across heterogeneous landscapes. We believe this dataset is essential to management and conservation biology initiatives not only for SBW and boreal forests but also for providing a starting point for broader evolutionary and ecological studies of complex population dynamics. Furthermore, the knowledge, data collection, and curation framework we present here can be used to inform similar spatial temporal baseline studies of other outbreaking (e.g., mountain pine beetle, red-backed voles) and invasive species (e.g., spongy moth and emerald ash borer). The data are released under a CC BY 4.0 license.

KEYWORDS

balsam fir, budworm, Canada, connectivity, genotype, Maine, New Brunswick, outbreak, Quebec, spruce, temporal

DATA AVAILABILITY STATEMENT

The dataset is available as [Supporting Information](#) and is also available in Figshare at <https://doi.org/10.6084/m9.figshare.25545724>. Raw data are provided in GenBank under accession no. PRJNA1120418 at <https://www.ncbi.nlm.nih.gov/bioproject/PRJNA1120418>. Scripts used for data processing are available at https://bitbucket.org/jerlar73/fast-gbs_v2/wiki/Home.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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